

## 3D Coordinates Extension — Build the Panda (Beginner)

**Topic:** 3D Panda (x, y, z) · **Course:** 3D Coordinates · **Level:** Beginner (Extension, after Lesson 6) · **Time:** about 30 minutes

Build a standing panda in 3D — **white body, black ears, black eyes, black legs**. Every block needs **three** numbers (**x, y, z**): x **across**, y **up**, z **deeper** (forward).

- Stand on your **home spot** (red block) every run. Move, and the panda moves too.
- 🎨 White parts = **white concrete**. Black parts = **black wool**.

### 1 · Predict

Read the code. Circle one word.

```
place black block at (2, 5, 1)
place black block at (2, 6, 1)
place black block at (2, 7, 1)
```

**Only y changes. This is a leg climbing...** Circle one: across · up · deeper

```
place white block at (2, 8, 1)
place white block at (2, 8, 2)
place white block at (2, 8, 3)
```

**Only z changes. Circle one:** across · up · deeper

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## 2 · Spot the Bug 🐛

An ear sits **on top** of the head, one step up, at (2, 9, 1). The code below has only **two** numbers, so it lands flat on the ground.

```
# broken  
place black block at (2, 9)
```

**Fill the blank to fix it:**

```
place black block at (2, 9, __)
```

**Which number fills the blank? Circle one:** 1 · 3

**What was missing? Circle one:** the z number · the color

### 3 · Show Your Work 📷 🎥

Look at the 🐼 panda in the world. Walk around it. See the ears up top, the eye patches, and the two legs at the bottom.

**Build a small panda on your home spot.** Keep it easy:

White head: fill white blocks from (2, 5, 1) up to (5, 8, 1) — a small square.

Two black ears on top: black block at (2, 9, 1) and (5, 9, 1).

Two black eyes on the front: black block at (2, 7, 1) and (5, 7, 1).

Two black legs at the bottom: black block at (2, 4, 1) and (5, 4, 1).

💡 Tip: ears go **up** (bigger y). Legs go **down** (smaller y). Eyes only show on the **front** (z = 1).

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

**1 · Your code.** Scroll through it. Say what each part does.

**2 · Your build.** Point the camera at the panda. Name the parts.

Say these out loud, filling in the blanks:

Today I built \_\_\_\_\_.

I built it using this code: \_\_\_\_\_.

In this code I used \_\_\_\_\_.

The hardest part was \_\_\_\_\_.

That part was hard because \_\_\_\_\_.

The most fun part was \_\_\_\_\_.

Something new I learned was \_\_\_\_\_.

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Send this worksheet + your **one** video to teacher on KakaoTalk.